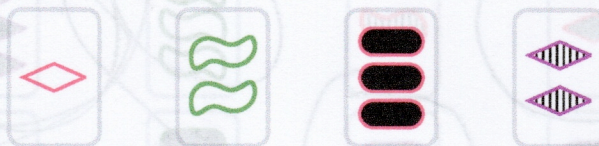


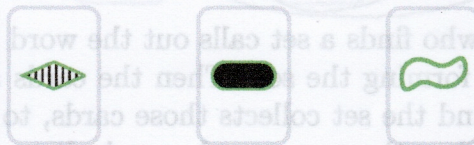
WORKSHEET 22

Set

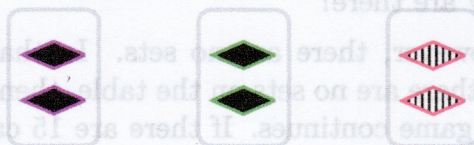
In this worksheet, we'll look at some of the mathematical properties of the popular game of SET. The game of SET is played with a deck of cards. Each card has four attributes: number, color, shape, and shading. There are three possibilities for each attribute. For number, the possibilities are 1, 2, and 3. For color, they are red, green, and purple. For shape, they are diamonds, ovals, and squiggles. For shading, they are hollow, shaded, and solid. Some sample cards are



A *set* consists of three cards such that, in every attribute, all the cards are either all the same or all different. For instance, the cards



form a set. They all have the same number and the same color, but they all have different shape and shading. On the other hand, the cards



do not form a set, because two of them are solid and one is shaded.

PROBLEM 22.1. In a SET deck, there is exactly one card with each type for each attribute. How many cards does the deck contain?

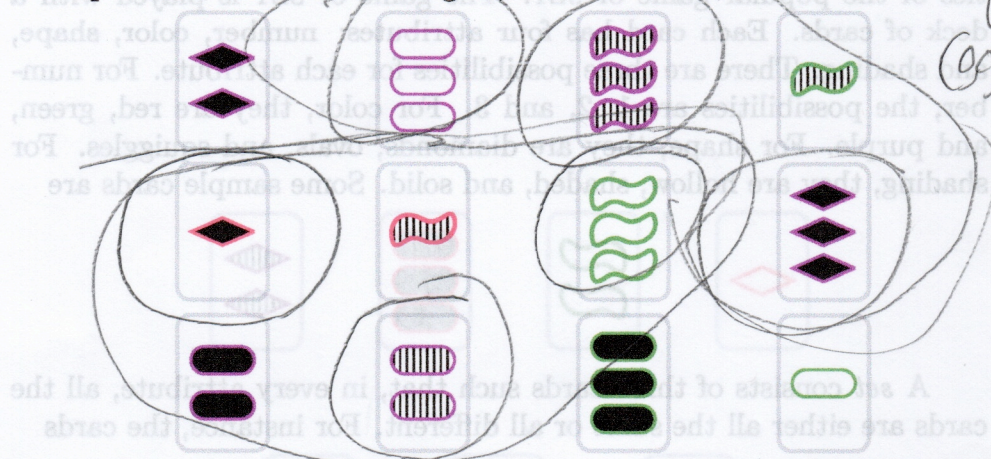
PROBLEM 22.2. Show that, for any two cards, there is exactly one card in the deck forming a set with the first two.

PROBLEM 22.3. What is the total number of sets?

PROBLEM 22.4. We can further subdivide based on the number of attributes where all three cards are the same. How many sets are there are 0, 1, 2, and 3 common attributes?

PROBLEM 22.5. We can represent a SET card as an ordered quadruple (a, b, c, d) , where each a, b, c, d is 1, 2, or 3. The first coordinate represents number, the second coordinate color, the third coordinate shape, and the fourth coordinate shading. If we use the ordering red, green, purple for color, diamond, oval, squiggle for shape, hollow, shaded, and solid for shading, then the quadruple $(1, 1, 2, 3)$ represents the card with one red solid oval. Suppose we have three cards represented by (a_1, b_1, c_1, d_1) , (a_2, b_2, c_2, d_2) , and (a_3, b_3, c_3, d_3) . What is the condition on these numbers that guarantees that they form a set?

Here is how to play SET as a game. First, twelve cards are dealt face up on the table in a 3×4 grid, like this:



Then, anyone who finds a set calls out the word "set!" and points to the three cards forming the set. Then the cards are removed, and the player who found the set collects those cards, to get points at the end of the game. Then three new cards are dealt to replace that set.

PROBLEM 22.6. Find a set in the arrangement above. Is there more than 1? How many are there?

Sometimes, however, there are no sets. In that case, when the players agree that there are no sets on the table, then three more cards are dealt, and the game continues. If there are 15 cards on the table, then the players try to find two sets before more cards are dealt, if possible, so that by default there are 12 cards on the table at any time. However, the number can dip below 12 if there are no more cards left in the deck.

PROBLEM 22.7. Find an arrangement of 12 cards with no set. Can you find more than 12 cards with no set? What is the maximum number?

PROBLEM 22.8. A common convention is to deal the last card in the deck face down. It can still be used as part of a set, but the players must determine what the card would be. Without memorizing all the cards that have been removed, how can they determine the last card?

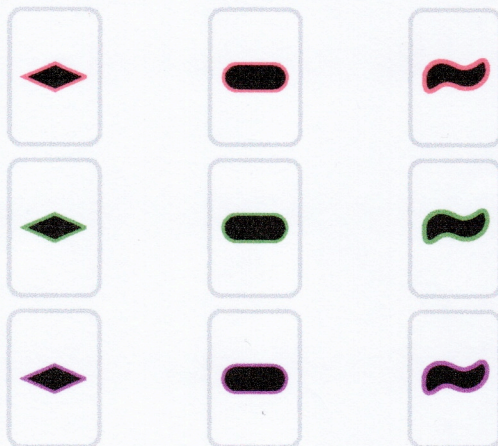
When 12 cards are left all other sets are taken and all cards sum is mod 3 = 0. So, if you see 3 sets you can get the 4th because mod 3.

Condition is that all three cards, when added by respect column, equals 0 in mod 3. (0,0,0)

16 cards no set: only two of each attribute present. 20 research?

PROBLEM 22.9. Suppose at the end of the game, there are three cards left on the table. Must they form a set?

Suppose we restrict to just two attributes, so we have 9 cards. For example, if we fix the shading as solid and number as 1, and vary only color and shape, we have the following 9 cards:



PROBLEM 22.10. How many ways are there to partition up these nine cards into three disjoint "parallel" sets?

PROBLEM 22.11. Are there other collections of 9 cards for which there are the same number of ways of partitioning them into three disjoint parallel sets? Of course, the answer is yes, because we can (for instance) make 2 of each shape, or make them all hollow, or various other things. But are there any substantially different such collections? (Figure out first what you think that should mean.)

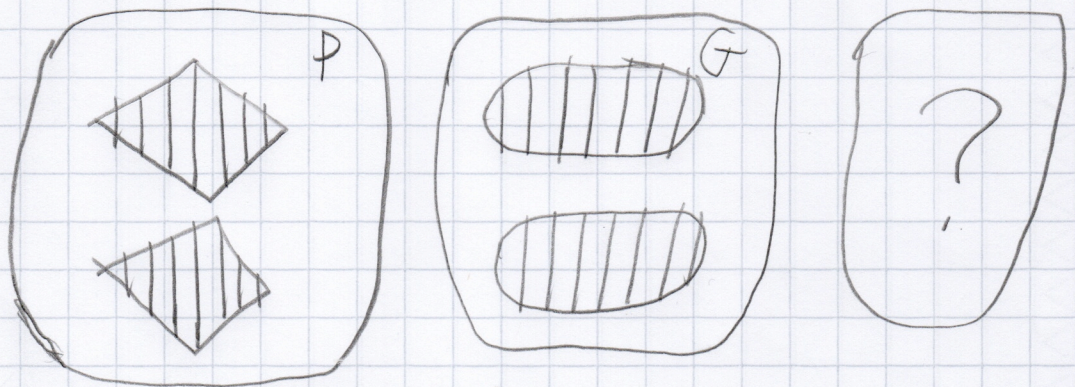
Actually no, show that any # of swaps or arrangement of either fixed or changeable attributes will get any 9-card ~~set~~ draw with 2 partitions into 3 sets back to the ↑ picture.

All sets taken
 $\equiv 0 \pmod 3$.
 Show that
 81 card deck
 $\equiv 0 \pmod 3$,
 then subtract
 so remaining
 3 cards = set.

Rows, columns,
 diagonals.
 only rows/
 columns for 3.

22.1: Because there are 4 attributes each with 3 possible choices, each card has $3 \cdot 3 \cdot 3 \cdot 3 = 81$ possible values, so 81 cards in the deck.

22.2: A pair of cards are either similar or different for each of the 4 attributes.



For example, these two cards are similar in Stripes and 2 but are different in Color (missing R) and Shape (Missing $\sim$). The only card that would complete this set is different than the pair's differences and similar to the pair's similarities - in this case, the only card is .

22.3: Because any pair can already form a set with only one other card, there are 81×80 pairs possible in the deck, which becomes 6480 sets. Not so fast, though! Order doesn't matter, so dividing by $3! = 6$ makes only 1080 possible sets in the deck.

22.4: 0 common attributes: Requires a pair of completely different cards to form. $81 \times (2^4 \text{ removing 1 attribute option each} = 16) = 1296$ sets, then $\div 6$ for ordering = 216 sets.

1 common attribute: Requires a pair of completely different except for one attribute cards to form. $\binom{4}{1} = 4$ ways to choose similar attribute, 3 choices of similarity. $3! = 6$ ways to arrange each of the different attributes, so $4 \times 3 \times 6 \times 6 \times 6 = 2592$, $\div 6$ for overcounting = 432 sets.

22.4 (cont.): 3 common attributes;

$\binom{4}{3} = 4$ ways to select similarities.

3 ways to select each similarity.

$3! = 6$ ways to arrange the difference.

$4 \cdot 3 \cdot 3 \cdot 3 \cdot 6 = 648$, $\div 6$ for overcounting = 108.

2 common attributes: Whatever's left over, or $1080 - 108 - 432 - 216 = 324$.

22.5: Add the quadruples together, by adding each corresponding

term. If the result is $(0, 0, 0, 0)$ in mod 3, then a set is present.

All different attributes

are 1, 2, and 3 which add to 6,

and all same attributes are

1, 1, 1, 2, 2, 2, or 3, 3, 3 which

add to 3, 6, and 9, respectively.

All of these sums are $0 \pmod 3$.

22.6: I found 2 total sets:

Shape				and				All
Shading	None	Diagonal lines	Diagonal lines		Diagonal lines	Diagonal lines	Diagonal lines	Diff.
Diff.	None	Diagonal lines	Diagonal lines		Diagonal lines	Diagonal lines	Diagonal lines	
Color	None	Diagonal lines	Diagonal lines		Diagonal lines	Diagonal lines	Diagonal lines	
Number	None	Diagonal lines	Diagonal lines		Diagonal lines	Diagonal lines	Diagonal lines	
Same.	$(3, 3, 2, 1)$	$(3, 3, 3, 2)$	$(3, 3, 1, 3)$		$(1, 1, 1, 3)$	$(2, 3, 3, 2)$	$(3, 2, 3, 1)$	

22.7: An easy way to make 16 cards with no set involves removing a 3rd choice for each option. Now, each attribute can only be 1 or 2, summing to only 4, 5 when different. No "all same" sets can exist, so there are many 16-card arrangements with no set. There are 20 card no-set arrangements, which are created by adding 4 cards one-by-one to the 16 cards, checking that no set exists with those cards.

22.8: If the last 12 cards are dealt, then therefore all other cards must have been taken as a set. So, the 12 cards left add to $(0,0,0,0)$ in mod 3.

The 2 cards with no set face up are then a hidden card set.

* (deck = 0 mod 3) - (taken = 0 mod 3) ^{# No other initial set of a grouping exists other than row/column/diagonal}

22.9: If 3 cards are left, all other cards taken must have been a set, so their vector sum is 0 mod 3.

The entire deck is also 0 mod 3 in vector sum because each attribute is partitioned evenly about the deck (27 1's, 27 2's, and 27 3's add to 0 mod 3.) Therefore, by subtraction* the remaining 3 cards are 0 mod 3 in vector sum, or a set.

22.10: There are 4 ways to partition these 9 cards: one by rows, one by columns, and two using each diagonal separately as a set.

22.11: Excluding the changing of assigned attributes and the swapping of card order, NO other 9-card 4-way arrangement exists.†